

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 4: Processor Fundamentals

Subtopic 4.1: Central Processing Unit (CPU) Architecture

Concept 4.1.7: Fetch-Execute Cycle and Register Transfer Notation

Total Questions: 10

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_11 (Q6.a)

- 6 (a) The processor uses several registers, including the Accumulator (ACC) and the Current Instruction Register (CIR).

Complete the table by describing the role of each register.

Register	Role
ACC	
CIR	

[2]

Question 2 | Source: 9618_w25_qp_12 (Q1.a)

- 1 Draw **one** line from each verification method to indicate whether it is used during data transfer or data entry.

Verification Method

Parity byte check

Checksum

Visual check

Parity block check

Use

Data transfer

Data entry

[2]

Question 3 | Source: 9618_w25_qp_12 (Q1.b)

- 1 Draw **one** line from each verification method to indicate whether it is used during data transfer or data entry.

Verification Method	Use
Parity byte check	
Checksum	Data transfer
Visual check	
Parity block check	Data entry

[2]

Question 4 | Source: 9618_w25_qp_12 (Q3.a.ii)

- (ii) Registers such as the Accumulator (ACC) and the Index Register (IX) are used in the CPU.

Identify **two** special purpose registers used in the CPU. Do **not** include the ACC or IX in your answers.

1

2

[2]

Question 6 | Source: 9618_s25_qp_13 (Q2.a.ii)

- 2 A computer has a processor.

(a) The processor has a Control Unit (CU) and system clock.

- (ii) The processor runs the Fetch-Execute (F-E) cycle.

Write the stages of the F-E cycle using register transfer notation.

.....

.....

.....

.....

.....

..... [4]

Question 7 | Source: 9618_w23_qp_13 (Q9.b)

9 A computer system is designed using the basic Von Neumann model.

Registers and buses are components in the Von Neumann model.

(b) The following incomplete table contains steps of the Fetch-Execute (F-E) cycle and their descriptions.

Complete the table by writing the missing steps using register transfer notation **and** the missing descriptions.

Step	Description
.....	The address in PC is incremented.
$MDR \leftarrow [MAR]$	
$MAR \leftarrow [PC]$	
.....	The contents of MDR are copied into CIR.

[4]

Question 8 | Source: 9618_s23_qp_13 (Q7.c)

7 A computer stores data in binary form.

(c) Complete the table by writing the register transfer notation for each stage of the Fetch-Execute (F-E) cycle given in the table.

Stage description	Register transfer notation
The Program Counter (PC) is incremented	
The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	

[2]

Question 9 | Source: 9618_w22_qp_12 (Q7.c)

- 7 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
STO	<address>	Store the contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
ADD	#n	Add the denary number n to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
JMP	<address>	Jump to the given address
CMP	<address>	Compare the contents of ACC with the contents of <address>
CMI	<address>	Indirect addressing. The address to be used is at the given address. Compare the contents of ACC with the contents of this second address
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
END		Return control to the operating system
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

- (c) Write the register transfer notation for each of the stages in the fetch-execute cycle described in the table.

Description	Register transfer notation
Copy the address of the next instruction into the Memory Address Register.	
Increment the Program Counter.	
Copy the contents of the Memory Data Register into the Current Instruction Register.	

[3]

Question 10 | Source: 9618_w21_qp_11 (Q6.a)

- 6 (a) There are **two** errors in the following register transfer notation for the fetch-execute cycle.

- 1 $MAR \leftarrow [PC]$
- 2 $PC \leftarrow [PC] - 1$
- 3 $MDR \leftarrow [MAR]$
- 4 $CIR \leftarrow [MDR]$

Complete the following table by:

- identifying the line number of each error
- describing the error
- writing the correct statement.

Line number	Description of the error	Correct statement

[4]

Question 11 | Source: 9618_s21_qp_11 (Q3.a)

- 3 A processor has one general purpose register, the Accumulator (ACC), and several special purpose registers.

- (a) Complete the following description of the role of the registers in the fetch-execute cycle by writing the missing registers.

The holds the address of the next instruction to be loaded. This address is sent to the

The holds the data fetched from this address.

This data is sent to the and the Control Unit decodes the instruction's opcode.

The is incremented.

[5]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q6.a)

6(a)	1 mark for each correct description, max 2 marks	2						
<table><tr><th>Register</th><th>Role</th></tr><tr><td>ACC</td><td>stores the intermediate results of arithmetic and logical operations // holds the result of a calculation</td></tr><tr><td>CIR</td><td>holds the instruction currently being decoded and/or executed</td></tr></table>			Register	Role	ACC	stores the intermediate results of arithmetic and logical operations // holds the result of a calculation	CIR	holds the instruction currently being decoded and/or executed
Register	Role							
ACC	stores the intermediate results of arithmetic and logical operations // holds the result of a calculation							
CIR	holds the instruction currently being decoded and/or executed							

Mark Scheme for Question 4 | Source: 9618_w25_ms_12 (Q3.a.ii)

3(a)(ii)	1 mark per bullet point, max 2 marks	2
	• Program counter (PC) • Memory Data Register (MDR) • Memory Address Register (MAR) • Current Instruction Register (CIR) • Status Register	

Mark Scheme for Question 5 | Source: 9618_w25_ms_13 (Q6.b)

6(b)	1 mark for each bullet point, max 4 marks	4
	• MAR stores the <u>address</u> of the next instruction / data to be read from / or written to memory • The address is received from the Program Counter (PC) • The MDR stores the data / instruction in the address stored in the MAR which has been read / written • The instruction passes to the CIR for decoding and executing	

Mark Scheme for Question 6 | Source: 9618_s25_ms_13 (Q2.a.ii)

2(a)(ii)	1 mark each • $MAR \leftarrow [PC]$ and $PC \leftarrow [PC] + 1$ • $MDR \leftarrow [[MAR]]$ • $CIR \leftarrow [MDR]$ • Correct order • For example: • $MAR \leftarrow [PC]$ • $PC \leftarrow [PC] + 1$ • $MDR \leftarrow [[MAR]]$ • $CIR \leftarrow [MDR]$	4
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Mark Scheme for Question 7 | Source: 9618_w23_ms_13 (Q9.b)

9(b)

1 mark for each correct answer (shaded cells)

4

Step	Description
$PC \leftarrow [PC] + 1$	The address in PC is incremented.
$MDR \leftarrow [[MAR]]$	The data in the location pointed to by the MAR is copied to the MDR.
$MAR \leftarrow [PC]$	The contents of PC are copied to the MAR.
$CIR \leftarrow [MDR]$	The contents of MDR are copied into CIR.

Mark Scheme for Question 8 | Source: 9618_s23_ms_13 (Q7.c)

7(c)	1 mark for each Register transfer notation	2						
<table><tr><th>Stage description</th><th>Register transfer notation</th></tr><tr><td>The Program Counter (PC) is incremented</td><td>$PC \leftarrow [PC] + 1$</td></tr><tr><td>The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)</td><td>$MDR \leftarrow [[MAR]]$</td></tr></table>			Stage description	Register transfer notation	The Program Counter (PC) is incremented	$PC \leftarrow [PC] + 1$	The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	$MDR \leftarrow [[MAR]]$
Stage description	Register transfer notation							
The Program Counter (PC) is incremented	$PC \leftarrow [PC] + 1$							
The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	$MDR \leftarrow [[MAR]]$							

Mark Scheme for Question 9 | Source: 9618_w22_ms_12 (Q7.c)

7(c)

1 mark for each correct row:

3

Description	Register transfer notation
Copy the address of the next instruction into the Memory Address Register.	$MAR \leftarrow [PC]$
Increment the Program Counter.	$PC \leftarrow [PC] + 1$
Copy the contents of the Memory Data Register into the Current Instruction Register.	$CIR \leftarrow [MDR]$

Mark Scheme for Question 10 | Source: 9618_w21_ms_11 (Q6.a)

6(a)	1 mark for identification of line and description of error 1 mark for the correct statement	4									
	<table> <tr> <th>Line number</th><th>Description of the error</th><th>Correct statement</th></tr> <tr> <td>2</td><td>Program Counter should be incremented, not decremented</td><td>$PC \leftarrow [PC] + 1$</td></tr> <tr> <td>3</td><td>It should be the contents of the address in the MAR</td><td>$MDR \leftarrow [[MAR]]$</td></tr> </table>	Line number	Description of the error	Correct statement	2	Program Counter should be incremented, not decremented	$PC \leftarrow [PC] + 1$	3	It should be the contents of the address in the MAR	$MDR \leftarrow [[MAR]]$	
Line number	Description of the error	Correct statement									
2	Program Counter should be incremented, not decremented	$PC \leftarrow [PC] + 1$									
3	It should be the contents of the address in the MAR	$MDR \leftarrow [[MAR]]$									

Mark Scheme for Question 11 | Source: 9618_s21_ms_11 (Q3.a)

3(a)	1 mark for each completed statement The Program Counter holds the address of the next instruction to be loaded. This address is sent to the Memory Address Register . The Memory Data Register holds the data fetched from this address. This data is sent to the Current Instruction Register and the Control Unit decodes the instruction's opcode. The Program Counter is incremented.	5
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